

Setting up for the lab

1. Restart the SQL Server service (clears stats)
2. Restore your StackOverflow database
3. Copy & run the Lab 6 setup script
4. If you saved your Lab 5 changes, apply those
5. Start SQLQueryStress:
 1. File Explorer, D:\Labs, run SQLQueryStress.exe
 2. Click File, Open, D:\Labs\ServerLab5.json
 3. Click Go

3.5 p1

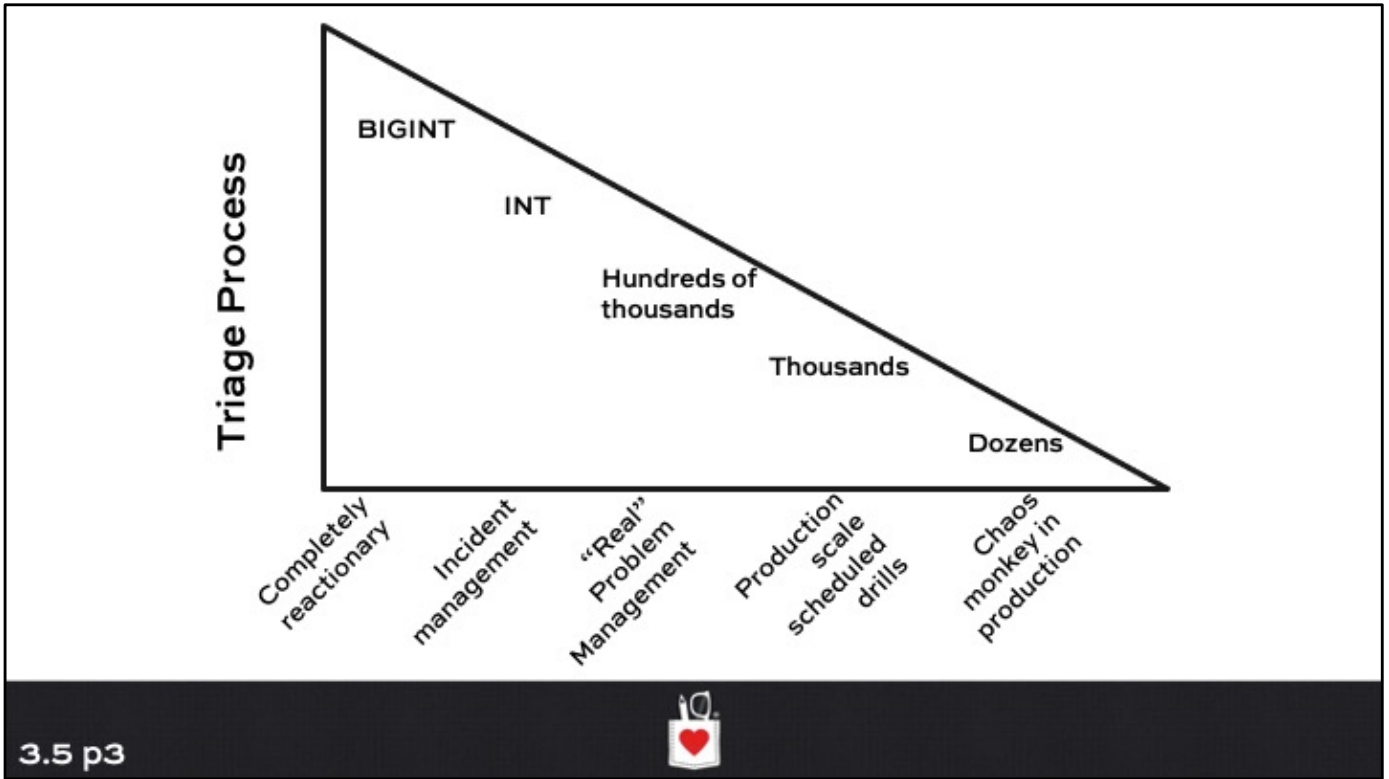




BRENT OZAR
UNLIMITED®

How to Triage SQL Server Emergencies

3.5 p2



Serious example: ITIL

Information Technology Infrastructure Library

Related areas:

- Incident Management
 - Major incident specializations
- Problem management
- Change management

It's a library – it's pretty dry

Take the pieces that work best for you

BrentOzar.com/go/itil

3.5 p4



Your goal: get one step better.

Start to build and document processes that are:

- Proven
- Repeatable
- Teachable – to help turn our juniors into seniors



Let's start with the
most common IT issue:
the Emergency Room.



Triage

is the process of finding out where the patient belongs.



Hospital triage levels

1. The patient is dead,
and we have to try to resuscitate them right now
2. The patient is likely to die regardless of whether
they receive medical help
3. The patient will live, but only if we help them now
4. The patient will live whether or not we help them



SQL Server performance triage

1. The server is down, not responding to queries
2. The server is going to go down in minutes if we can't fix the problem quickly
3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



SQL Server performance triage

1. The server is down, not responding to queries
2. The server is going to go down in minutes if we can't fix the problem quickly

**At these levels, you may have to
take risks and perform drastic actions.
You need to know which doctors to bring into the room.**



SQL Server performance triage

At these levels, taking risks or drastic actions can easily make things worse.

We need to prove it's not an emergency and respond appropriately.

4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow



SQL Server performance triage

Sometimes Bob is just faking it.
That doesn't mean we ignore him, we just set priorities.

6. Bob's report is running just like it did yesterday: slow



Your goal is to fill this out fast.

The symptoms are _____.

I believe the root cause is _____.

If we don't take action now, the effect will be _____.

I recommend that we:

- Take action _____ now to fix it, or
- Take action _____ to investigate further, or
- Put this in _____'s queue

3.5 p13



Your triage tools, in order:

1. Your monitoring software
2. `sp_WhoIsActive`
3. `sp_BlitzFirst`
4. `sp_BlitzCache`
5. `sp_BlitzIndex`
6. `sp_Blitz`



The help desk can use these too.

The stored procedures require SA-level permissions.

You can grant permissions to procs rather than users.

It's crazy easy: <http://www.brentozar.com/askbrent/>



```
USE master;
GO
CREATE CERTIFICATE sp_BlitzFirst_cert
  ENCRYPTION BY PASSWORD = '50ClockSomewhere'
  WITH SUBJECT = 'Certificate for sp_BlitzFirst',
  START_DATE = '20130711', EXPIRY_DATE = '21000101';
GO
CREATE LOGIN sp_BlitzFirst_login FROM CERTIFICATE sp_BlitzFirst_cert;
GO
CREATE USER sp_BlitzFirst_login FROM CERTIFICATE sp_BlitzFirst_cert;
GO
GRANT EXECUTE ON dbo.sp_BlitzFirst TO sp_BlitzFirst_login;
GO
GRANT CONTROL SERVER TO sp_BlitzFirst_login;
GO
ADD SIGNATURE TO sp_BlitzFirst BY CERTIFICATE sp_BlitzFirst_cert
  WITH PASSWORD = '50ClockSomewhere';
GO
GRANT EXECUTE ON dbo.sp_BlitzFirst TO [public];
GO
```

3.5 p16



Your monitoring software

3.5 p17



What part does monitoring do?

The symptoms are ____.

I believe the root cause is ____.

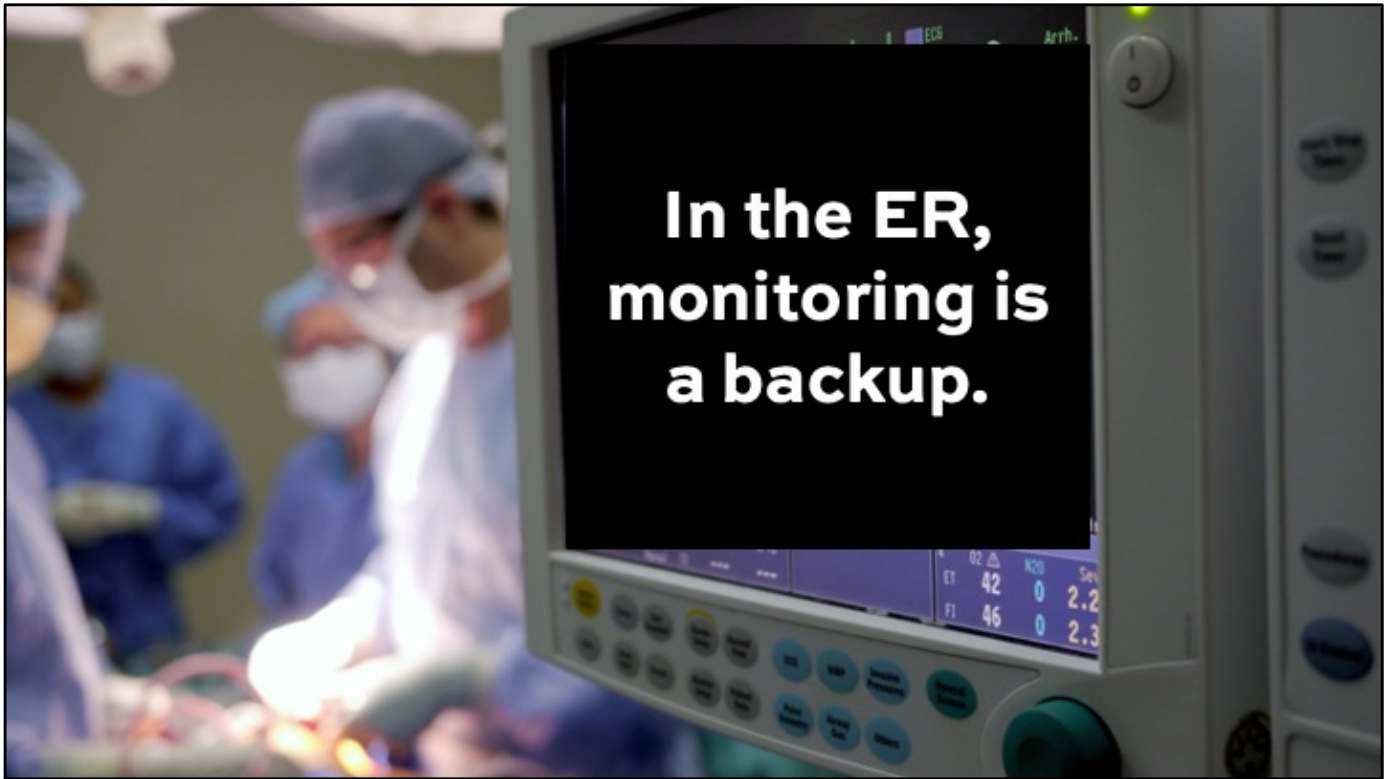
If we don't take action now, the effect will be ____.

I recommend that we:

- Take action ____ now to fix it, or
- Take action ____ to investigate further, or
- Put this in ____'s queue









So monitoring software only:

1. Emails you when something is really on fire
(and let's be honest, you have a rule that puts those emails in a folder anyway)
2. Helps when you need to look at the history of a specific metric, and compare it to today
(and you know what that metric means, and what it should be)





sp_WhoIsActive

3.5 p24



BrentOzar.com/go/active

Totally free diagnostic tool by Adam Machanic

Installs in the master database

Runs immediately, or in seconds

One-time point-in-time picture of running queries

Start with no parameters for the fastest picture:

```
EXEC dbo.sp_WhoIsActive
```



SQL Server performance triage

1. The server is down, not responding to queries

**If `sp_WhoIsActive` doesn't run,
you're not triaging performance.
It's an outage. Use our First Responder Kit:
BrentOzar.com/responder**



Triaging sp_WhoIsActive's output

No rows: nothing's going on.

Have any queries been running for hours?

Have any queries been running for minutes?

Is anyone being blocked?

Is someone using up all of TempDB?
(See the tempdb allocations column)





Triage is part 1, and only part 1.

1. **Identify the root cause of the problem.**
2. Propose a method for pain relief.
3. Make changes to relieve the pain.
4. Measure to prove that the pain is gone.
5. If it's not gone, back the change out, and go to 1.
6. If it's gone, document what happened.



Round two: getting more info

```
EXEC sp_WhoIsActive
```

```
@get_locks = 1, @get_plans = 1
```

Returns clickable XML fields with locks and plans



Checkpoint: what level are we?

1. The server is down, not responding to queries
2. The server is going to go down in minutes if we can't fix the problem quickly
3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



SQL Server performance triage

-
2. The server is going to go down in minutes if we can't fix the problem quickly

If there's a huge blocking chain,
you might be done right now.



Fill this out, get consensus, act.

The symptoms are a frozen web site and query timeouts.

I believe the root cause is a blocking chain led by Norm.

If we don't take action now, the effect will be continued blocking, log file growth, and the web site will be unusable.

I recommend that we:

- Take action to kill ~~Norm~~ Norm's query now to fix it, and
- Put this in the help desk queue to remove Norm's access



SQL Server performance triage

1. The server is down, not responding to queries
2. The server is going to go down in minutes if we can't fix the problem quickly

If `sp_WhoIsActive` runs, and doesn't show a huge blocking chain, you can probably rule out #1 and #2.



SQL Server performance triage

Now let's move on to these.

3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



sp_BlitzFirst

3.5 p36



BrentOzar.com/askbrent

Totally free diagnostic tool by an amazingly hot guy

Installs in the master database

Runs in 5 seconds ideally, but can be more under load

Snapshot of the server's bottlenecks right now

Start with ExpertMode on because you're smart:

```
EXEC dbo.sp_BlitzFirst @ExpertMode = 1
```



Telling: what isn't there

Backups, restores

PLE low

DBCCs

Slow reads or writes

Data or log file
growing or shrinking

Compilations/sec
high

Long-running query
blocking others

Plan cache erased
recently



Checkpoint: what level are we?



3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



Can we fill this out yet?

The symptoms are _____.

I believe the root cause is _____.

If we don't take action now, the effect will be _____.

I recommend that we:

- Take action _____ now to fix it, or
- Take action _____ to investigate further, or
- Put this in _____'s queue

3.5 p40



Fill this out, get consensus, act.

The symptoms are a frozen web site and query timeouts.

I believe the root cause is a surprise backup run by Norm the new guy.

If we don't take action now, the effect will be continued slow query response times for about 30 minutes.

I recommend that we:

- Put this in the help desk's queue to coordinate questions between the end user and the lead developer



Checkpoint: what level are we?

sp_BlitzFirst might rule out #3,

but if #3 is happening, sp_BlitzFirst doesn't get you a root cause.

Either way, the next step is the same.

3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



sp_BlitzCache

3.5 p43



BrentOzar.com/blitzcache

Analyzes the top 10 most resource-intensive queries in the server's plan cache

Includes all databases

Runs in 10-60 seconds (or more) depending on server horsepower, memory size (doesn't block)



Is it the usual suspects?

Are they queries you recognize?

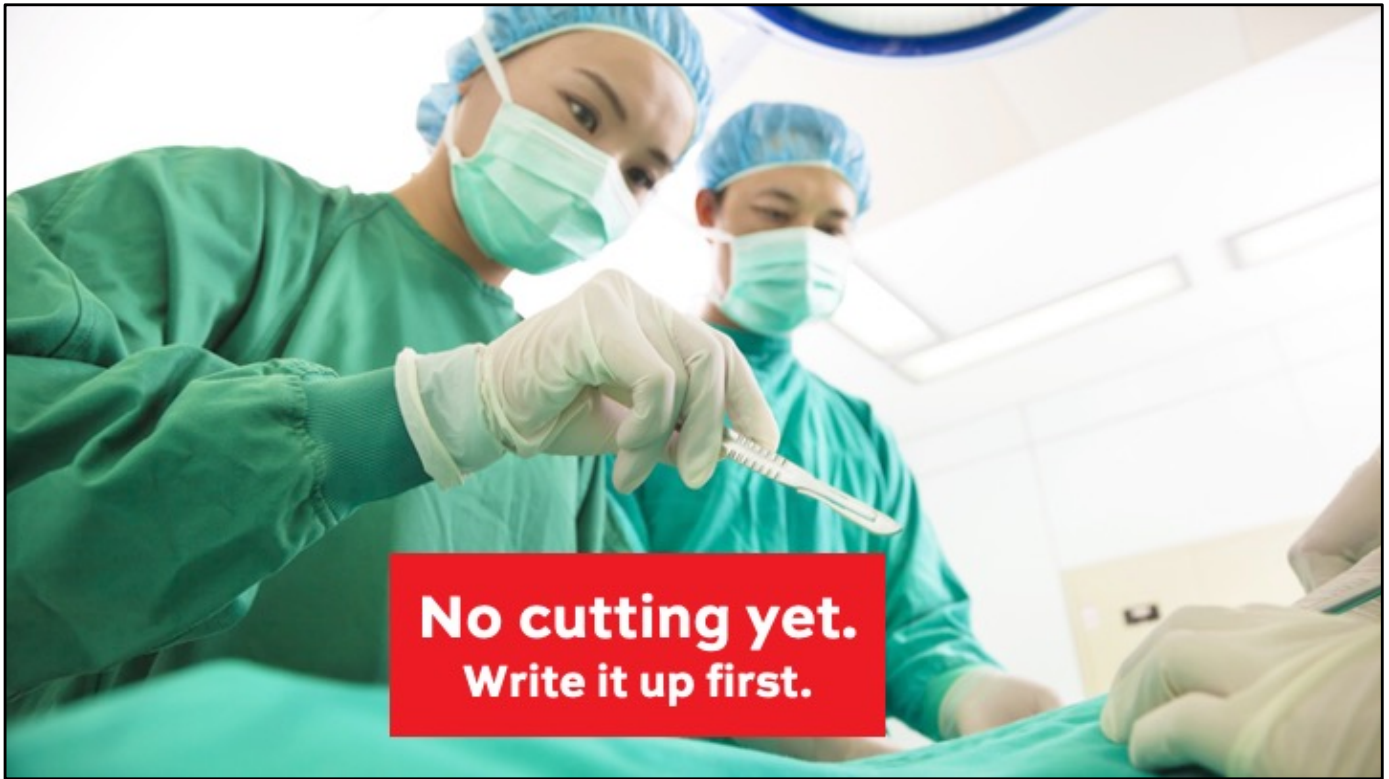
Are they using the same level of resources they usually use?

Is there 1 query suddenly using >50% of resources?

Did the query get compiled with unusual parameters?

We'll cover the mechanics of fixing it this week.





Can we fill this out yet?

The symptoms are _____.

I believe the root cause is _____.

If we don't take action now, the effect will be _____.

I recommend that we:

- Take action _____ now to fix it, or
- Take action _____ to investigate further, or
- Put this in _____'s queue



Fill this out, get consensus, act.

The symptoms are a frozen web site and query timeouts.

I believe the root cause is a bad query plan for
GetCommentsDetails.

If we don't take action now, the effect will be a frozen web
site and query timeouts for hours/days.

I recommend that we:

- Take action by freeing this plan from cache now, and
- Put this in the developers' queue to make
GetCommentsDetails less vulnerable to parameter
sniffing issues



Checkpoint: what level are we?

sp_BlitzCache identifies #3-#5.

Now it's just down to the stuff that has been slow for weeks.

3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



sp_BlitzIndex

3.5 p50



BrentOzar.com/blitzindex

Diagnoses index design issues
(not fragmentation)

By default, analyzes the current database

Use @GetAllDatabases = 1 to check 'em all

Max 50 databases – unless @BringThePain = 1



Can we fill this out yet?

The symptoms are _____.

I believe the root cause is _____.

If we don't take action now, the effect will be _____.

I recommend that we:

- Take action _____ now to fix it, or
- Take action _____ to investigate further, or
- Put this in _____'s queue



Fill this out, get consensus, act.

The symptoms are slow reports on the Sales table.

I believe the root cause is improper indexing on that table.

If we don't take action now, the effect will be continued slow queries, but no other parts of the app will be impacted.

I recommend that we:

- Take action by creating these 4 indexes on the development environment to investigate further, or
- Put this in my performance tuning queue



Checkpoint: what level are we?

sp_BlitzIndex helps you fix #3-#6,
but only if there's indexing problems. There may be bigger ones.

3. The entire server is in bad shape, but will survive today whether we fix the problem or not
4. Some parts of some applications are unusually slow
5. Bob's report is running unusually slow
6. Bob's report is running just like it did yesterday: slow



sp_Blitz

3.5 p55



BrentOzar.com/blitz

Totally free health check tool by a helpful idiot.

Runs in 10-60 seconds (or more) depending on server horsepower, user database count

Gives you a prioritized list of health problems

```
EXEC dbo.sp_Blitz @CheckServerInfo = 1
```



This is not an
emergency tool.



Triage recap

3.5 p58



Your goal is to fill this out.

The symptoms are _____.

I believe the root cause is _____.

If we don't take action now, the effect will be _____.

I recommend that we:

- Take action _____ now to fix it, or
- Take action _____ to investigate further, or
- Put this in _____'s queue



Your tools to triage performance

1. Your monitoring software
2. sp_WhoIsActive
3. sp_BlitzFirst
4. sp_BlitzCache
5. sp_BlitzIndex
6. sp_Blitz

